LOW-VOLTAGE DOMAIN (JLCPCB 4-LAYER: F.Cu, In1.Cu GND, In2.Cu Power, B.Cu)

01_power_mcu

VIN_PROTECTED
3V3
GND
DRV_IN1
DRV_IN2
INTERLOCK_OK
HV_SENSE_ADC
HV_CURR_ADC

File: 01_power_mcu.kicad_sch

04_sense_safety

VIN_PROTECTED
3V3
GND
INTERLOCK_OK
HV_SENSE_ADC
HV_CURR_ADC

File: 04_sense_safety.kicad_sch

02_drive_transformer

HV_VM
DRV_SLEEP
GND
DRV_IN1
DRV_IN2
HV_RAW
HV_RETURN

File: 02_drive_transformer.kicad_sch

HIGH-VOLTAGE MULTIPLIER DOMAIN (STRICT MULTI-LAYER KEEPOUT

03_multiplier

SEC_A
HV_RETURN
HV_RAW
HV_RETURN

File: 03_multiplier.kicad_sch

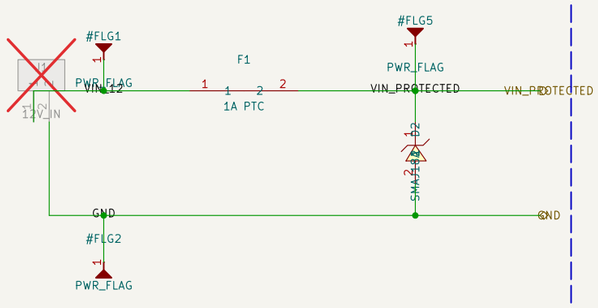
Antigravity Engineering

Sheet: /
File: ionic_thrust_hv.kicad_sch

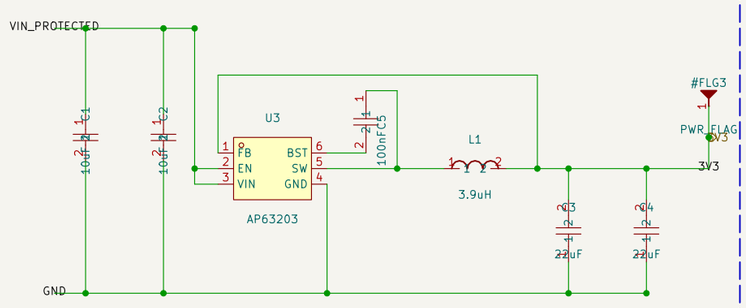
Title: ESP32 12V to 15kV DC Corona Supply — System Overview

Size: A3	Date: 2026-10-01	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/5

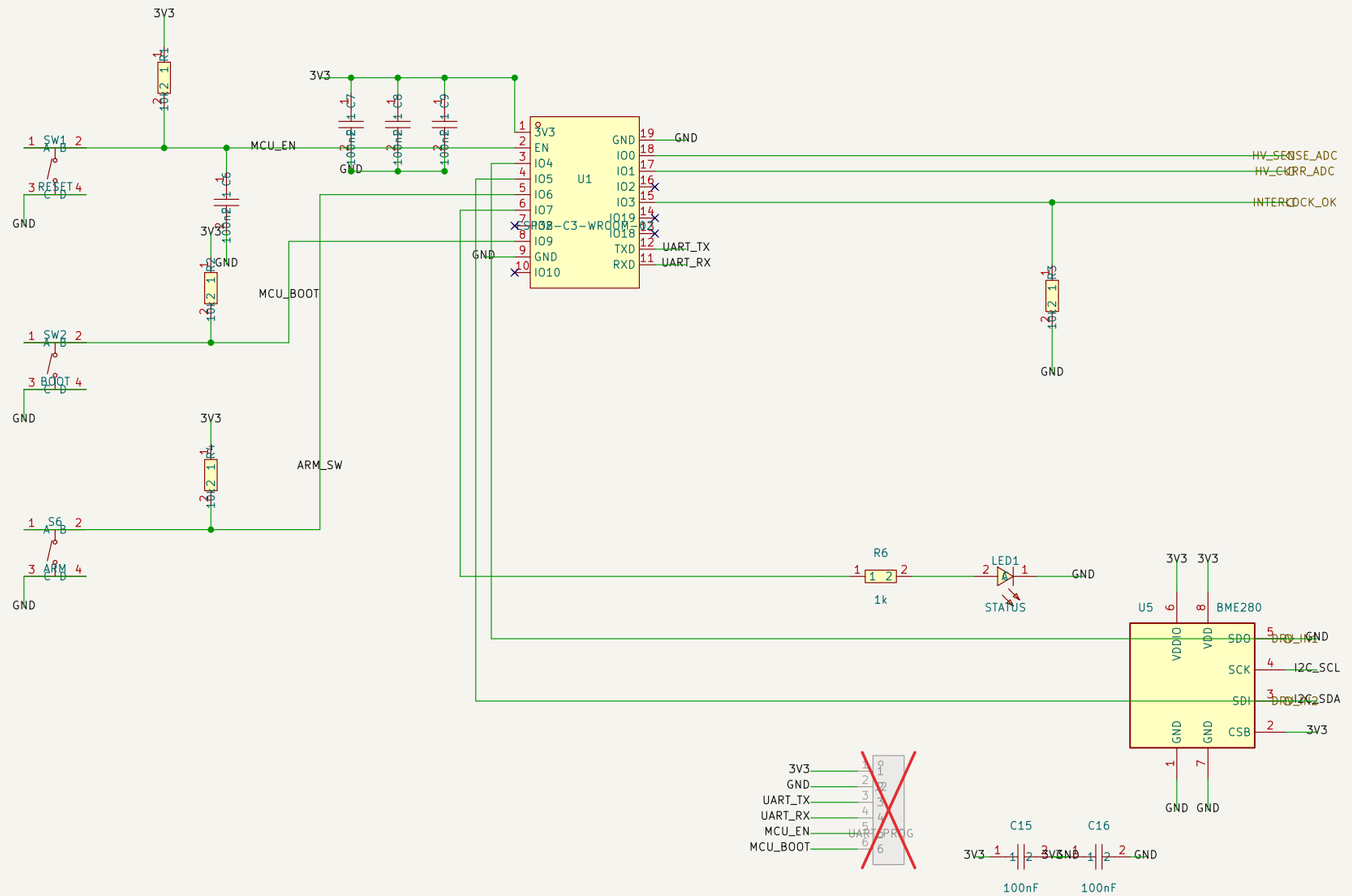
12V DC INPUT & OVERVOLTAGE PROTECTION



3.3V SYNCHRONOUS BUCK REGULATOR (AP63203WU-7)



ESP32-C3 CONTROLLER & USER INTERFACE



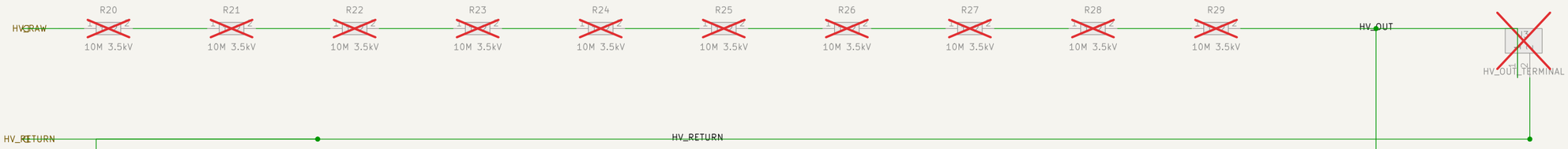
Antigravity Engineering

Sheet: /01_power_mcu/
File: 01_power_mcu.kicad_sch

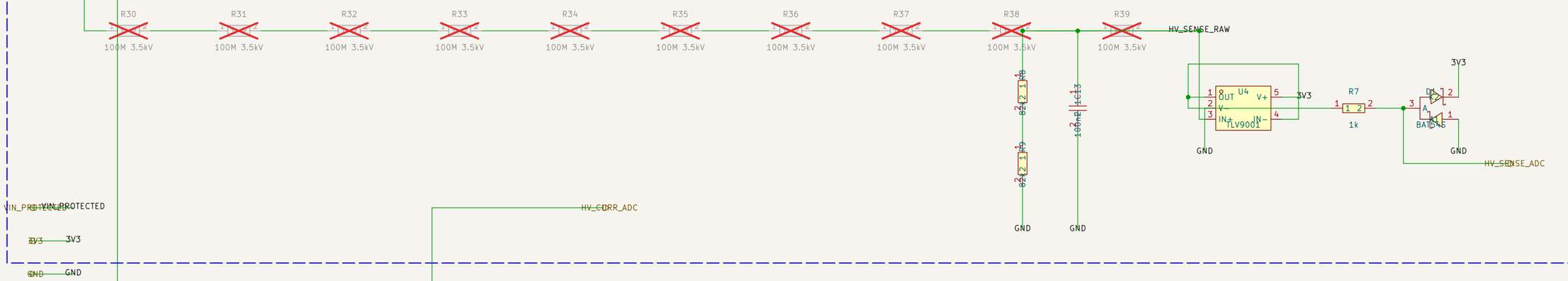
Title: 01 Power Supply & ESP32-C3 Microcontroller

Size: A3	Date: 2026-10-01	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 2/5

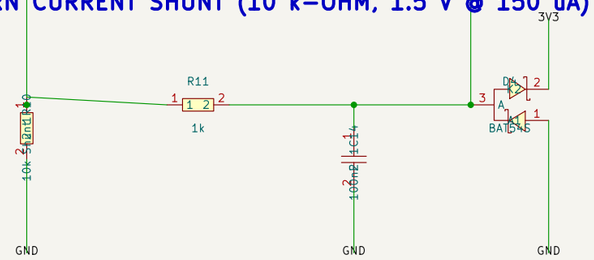
OUTPUT BALLAST STRING (100 M-OHM TOTAL, 150 uA SHORT-CIRCUIT LIMIT)



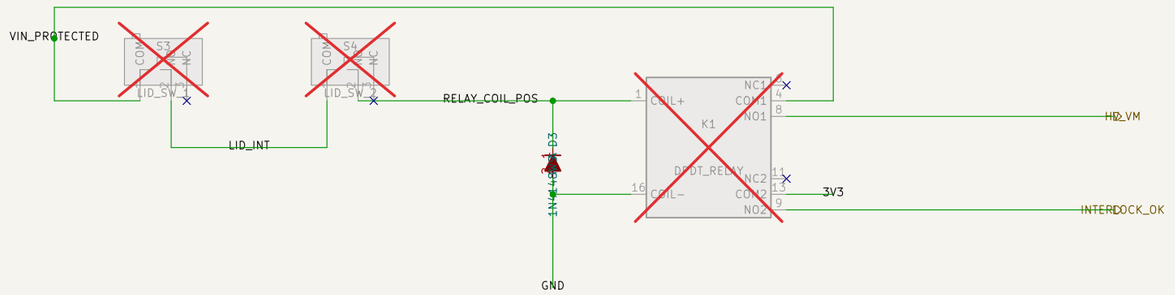
1 G-OHM HIGH VOLTAGE DIVIDER (RATIO 6098:1) & BUFFER



HV RETURN CURRENT SHUNT (10 k-OHM, 1.5 V @ 150 uA)

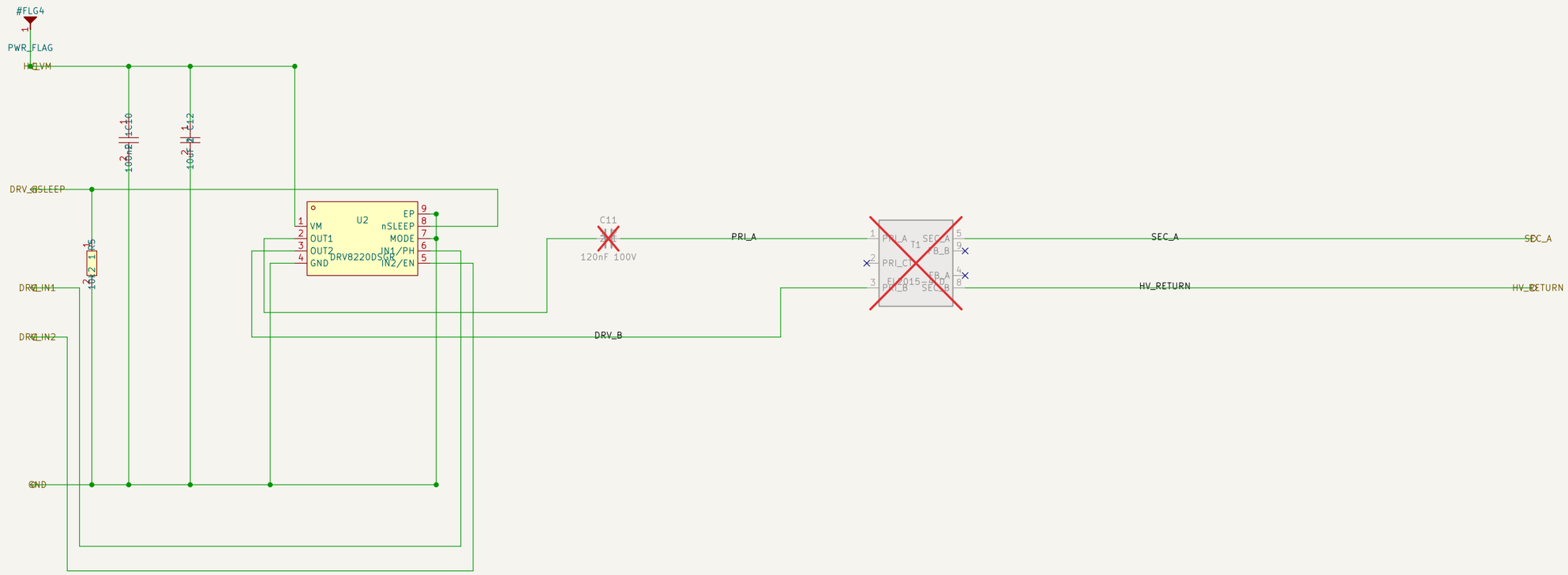


HARDWARE SAFETY INTERLOCK (DUAL REDUNDANT LID SWITCHES)



Antigravity Engineering		
Sheet: /04_sense_safety/		
File: 04_sense_safety.kicad_sch		
Title: 04 Ballast, Divider, Sensing & Hardware Interlock		
Size: A3	Date: 2026-10-01	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 3/5

FULL-BRIDGE DRIVER & SERIES RESONANT STEP-UP TRANSFORMER

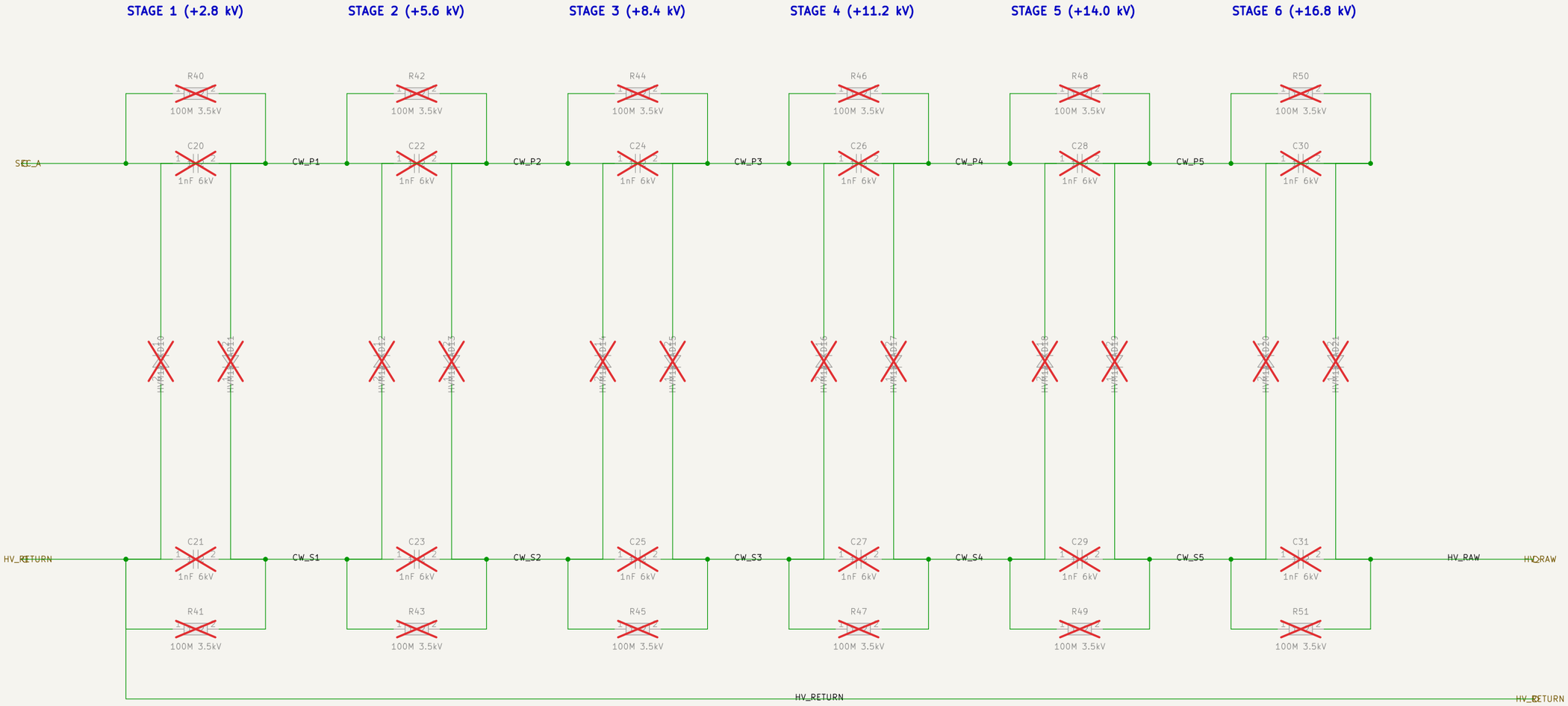


RESONANT TANK SPECIFICATIONS:

1. Resonant frequency fr ≈ 69.3 kHz (L_pri ≈ 44 uH, C11 = 120 nF 100V film).
2. Transformer ratio: ~1:100 step-up, delivering ~1.4 kV AC peak to Cockcroft-Walton ladder.
3. HV_RETURN is isolated from low-voltage GND; connects via 10k current shunt on Sheet 4.

Antigravity Engineering		
Sheet: /02_drive_transformer/		
File: 02_drive_transformer.kicad_sch		
Title: 02 Isolated AC Generation & Resonance Tank		
Size: A3	Date: 2026-10-01	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 4/5

6-STAGE POSITIVE COCKCROFT-WALTON VOLTAGE MULTIPLIER & PASSIVE BLEEDERS



SAFETY & DISCHARGE SPECIFICATIONS:

1. Nominal open-circuit output voltage: ~17 kV DC (regulated by MCU PWM duty cycle to +10 kV .. +15 kV DC).
2. PASSIVE DISCHARGE: 12x 100M 3.5kV bleeder resistors discharge all capacitors to < 50V within 5 seconds of shutdown.
3. CLEARANCE: Minimum 4.0 mm clearance maintained between adjacent stages; all inner copper planes removed beneath ladder.

Antigravity Engineering		
Sheet: /03_multiplier/		
File: 03_multiplier.kicad_sch		
Title: 03 6-Stage Positive Cockcroft-Walton Multiplier & Bleeders		
Size: A3	Date: 2026-10-01	Rev: B
KiCad E.D.A. 10.0.6		Id: 5/5